Map & Routing System: Architecture Analysis & Engineering Proposal

Target System: Nueva Ecija Travel Itinerary App ("Trip mo, trip ko") • Live Deployment: edselrenz.github.io/Itinerary-App/tabs/home • Date: November 6, 2026

1. Executive Summary: Live Verification Exposes Four High-Impact Map & Routing Gaps

A combined static code audit of the uploaded monorepo (`Project/` built on Angular 22, Ionic 9, and Capacitor 8; `backend/` built on FastAPI and SQLAlchemy 2.0) and interactive browser testing against the live deployment at https://edselrenz.github.io/Itinerary-App/tabs/home evaluated how both **geospatial mapping/routing** and **client-side SPA navigation** operate in production.

While client-side route transitions via `@angular/router` load cleanly across deep-linked views without page reloads, the spatial and routing architecture exhibits four critical-to-high engineering gaps that prevent travelers from visually planning or following multi-stop trips:

- **Silent Map Pin Rendering Failure on Production Builds (`MAP-01`, Critical):** Navigating to https://edselrenz.github.io/Itinerary-App/tabs/map or deep-linking via View Pin on Map (Rohi Paperflower Cafe) renders OpenStreetMap raster tiles and reports `13 destinations` in the drawer pill, but renders **zero marker pins** on the map canvas due to an ES module namespace / UMD global initialization bug with `leaflet.markercluster`.
- **Zero In-App Route Polylines or Multi-Stop Directions (`ROUTE-01`, High):** Generated and custom itineraries (/suggested-wizard, /custom-builder, and saved trips under /tabs/history) render purely as vertical text cards with no route map, no connecting polylines, no leg-by-leg travel distances or durations, and single-point external Google Maps links that drop intermediate waypoints.
- **176.2 km Cross-Province Backtracking in Suggested Presets & Chaining (`ROUTE-02`, High):** Offline day presets in `offline-places.data.ts` route travelers from General Tinio to Pantabangan (`57.2 km` north) and back to Gapan City (`58.9 km` south) in a single afternoon, while backend post-selection chronological sorting (`DEF-03`) reorders stops by time slot without re-optimizing spatial sequence.
- **Truncated Client-Side Post-Filtering & Hardcoded Coordinate Fallbacks (`MAP-02`, `MAP-03`, `ROUTE-03`, High):** Municipality and price filters in `MapPage` filter only the first 100/150 rows in browser memory rather than querying the backend database; deep-linking with coordinates skips opening the target POI bottom sheet; and adding a Nearby Place injects hardcoded Bongabon coordinates (`15.63, 121.14`) regardless of the active municipality.

2. Current Architecture: How Routing and Mapping Work Today

The application separates navigation and spatial logic across three distinct layers: the interactive Leaflet discovery map, the deterministic itinerary clustering and external directions layer, and the Angular SPA router.

2.1 Interactive Map Subsystem (`MapPage` & `PlaceService`)

- **Map Canvas & Provincial Bounds:** `MapPage.initMap()` (`Project/src/app/tabs/map/map.page.ts:249–314`) initializes a Leaflet map centered on `NUEVA_ECIJA_CENTER = [15.58, 121.0]` (zoom `10`, range `9–18`) locked to `NUEVA_ECIJA_BOUNDS = [[15.05, 120.55], [16.25, 121.55]]` with `maxBoundsViscosity: 0.8`. Raster tiles load from `https://tile.openstreetmap.org/{z}/{x}/{y}.png` and are cached for offline use via Angular Service Worker (`Project/ngsw-config.json`, up to 1,500 tiles for 14 days).
- **Viewport & Filter Pipeline:** On initialization and debounced `moveend` events (`350ms` when `zoom ≥ 10`), `loadPlaces(bounds?)` calls `PlaceService.searchPlaces()` (`GET /api/v1/places`) passing `q`, `categories`, and optional bounding-box parameters (`min_lat`, `max_lat`, `min_lng`, `max_lng`, `limit: 150`). When the FastAPI backend is unreachable, `PlaceService` falls back to the 13 curated POIs in `CANONICAL_OFFLINE_PLACES` (`Project/src/app/core/data/offline-places.data.ts`).
- **Client-Side Post-Filtering & Drawer UI:** Once `searchPlaces()` returns, `applyLocalFilters()` (`map.page.ts:386–416`) filters the returned slice in memory by `selectedMunicipality` and `selectedPriceTier` (`all`, `free`, `budget ≤ ₱500`, `mid > ₱500`). Users can inspect results via the collapsible `List View` drawer or tap a marker to open `.poi-bottom-sheet`.

2.2 Geospatial Clustering, Commute Guide & External Directions

- **Backend Haversine Generator:** `generate_suggested_itinerary_data()` (`backend/app/algorithms/itinerary_generator.py:182–390`) resolves a starting coordinate anchor from the target municipality or place, pre-filters active POIs in SQL within a coordinate bounding box (`cluster_radius_km + 10.0 km`), and scores candidates using great-circle Haversine distance (`+30 max` within `25 km`), preference match (`+40`), featured status (`+20`), same-day category diversity (`-25`), and cumulative daily budget fit (`+10` / `-30`).
- **Greedy Anchor Chaining vs. Post-Selection Time Sort:** During slot selection (`09:00`, `12:00`, `14:30`), the generator updates `current_anchor_lat/lng` to the just-selected place. However, lines `331–351` then reassign time slots if an accommodation is present and re-sort items strictly by `planned_time` without checking whether the new chronological order introduces spatial backtracking.
- **Static Commute Guide & Keyless Google Maps Handoff:** `getCommuteInfoForMunicipality()` (`Project/src/app/core/data/commute-guide.data.ts:89–139`) returns static terminal descriptions for 8 municipalities (always assuming departure from *Cabanatuan City Central Terminal*). All direction links across the app open single-destination external URLs (`https://www.google.com/maps/dir/?api=1&destination={lat},{lng}`).

2.3 Client-Side SPA Router Topology (`@angular/router`)

- **Standalone Lazy-Loaded Routes:** Configured in `Project/src/main.ts` with `IonicRouteStrategy`, `withPreloading(PreloadAllModules)`, and `withComponentInputBinding()`. Top-level routes in `app.routes.ts` handle onboarding/auth (/welcome, `/login`, `/signup`, `/guest`) and full-screen itinerary workflows (/custom-builder, `/custom-builder/:id`, /suggested-wizard, `/itinerary/:id`), while `/tabs` (`tabs.routes.ts`) hosts the 4 primary tabs (/tabs/home, /tabs/map, /tabs/history, /tabs/profile).

3. Root-Cause Diagnostic Matrix: Bugs & Architectural Limitations

The table below documents all eight verified map and routing defects identified across source inspection and live browser execution on edselrenz.github.io/Itinerary-App/tabs/home, ordered by severity.

ID & Severity	Module & Line Reference	Observed Behavior & Root Cause	Engineering Remediation
MAP-01 Critical	`map.page.ts` L47–48, L137–139, L288–293, L422	**Zero map pins render in production.** Static ES imports `import * as L from 'leaflet'; import 'leaflet.markercluster';` execute before `MapPage.constructor()` sets `window.L = L`. Under Angular esbuild, `L` is a frozen ESM namespace object, so `L.markerClusterGroup` is undefined or fails in `initMap()`, leaving `this.markerCluster = null` and causing `updateMarkers()` to return early.	Resolve the canonical Leaflet object via `const leaflet = (L as any).default ?? L; (window as any).L = leaflet;` at module top-level before importing or invoking `markerClusterGroup`, and add a graceful fallback to a standard `L.layerGroup()` if cluster initialization fails.
MAP-02 High	`place.service.ts` L119–149; `map.page.ts` L340–416; `places.py` L92–163	**Truncated client-side filtering & ignored offline bounds.** `getFallbackPaginatedPlaces()` ignores `min_lat/max_lat/min_lng/max_lng` and `featured`. Online, `MapPage` fetches 100/150 items globally and applies municipality and price filters in memory (`applyLocalFilters`), yielding 0 matches if matching POIs sit past row 150 in PostgreSQL.	Add `municipality: str | None` and `max_cost: float | None` query parameters to `GET /api/v1/places` in `places.py`, pass them from `MapPage.loadPlaces()`, and enforce bounding-box, municipality, and price filtering inside `getFallbackPaginatedPlaces()`.

ID & Severity	Module & Line Reference	Observed Behavior & Root Cause	Engineering Remediation
MAP-03 High	`map.page.ts` L171–186, L320–329; `home.page.ts` L229–230, L434–447	**Deep-link pin selection skipped & fallback coordinates hardcoded.** In `map.page.ts:177`, when `lat` and `lng` query params exist, the `if/else if` skips `flyToDestinationOrMunicipality()`, so `selectedPlace` is never set and the bottom sheet stays closed. Also, `home.page.ts:229` hardcodes `15.48, 120.97` on offline search errors.	Pass `placeId` directly in `HomePage.viewDestinationOnMap()`, select and open the matching `Place` in `MapPage` even when `lat/lng` are provided, and suppress the `moveend` reload during programmatic `flyTo()` animations.
ROUTE-01 High	`itinerary-detail.page.ts`; `suggested-wizard.page.ts`; `custom-builder.page.ts`	**No itinerary route map, polylines, leg metrics, or multi-stop directions.** Travelers cannot visualize their daily stop sequence on a map, see road distances/travel times between stops, or launch a multi-stop route in Google Maps (only single-stop ? `destination=lat,lng` links exist).	Introduce a shared `RouteMapComponent` & `RoutingService` that renders numbered stop pins, road-network polylines (via OSRM / GeoJSON fallback), per-leg distance/time badges, a *View Day Route on Map* deep-link, and multi-stop Google Maps URLs (`&origin=...&destination=...&waypoints=...`).
ROUTE-02 High	`offline-places.data.ts` L446–482; `itinerary_generator.py` L258–354	**176.2 km cross-province zig-zag in presets & unoptimized stop order.** Offline Day 2 (`58.5 km`) and Day 3 (`116.2 km`: General Tinio → Pantabangan → Gapan City) cross the province twice. Online, `DEF-03` re-sorts stops chronologically after greedy chaining without optimizing route order.	Re-cluster offline presets into coherent corridors (East/Bongabon, North/San Jose–Pantabangan, South/Cabanatuan–Gapan–Gen. Tinio, cutting distance by `66.3%` to `59.4 km`) and apply a 2-opt Nearest-Neighbor TSP pass to activity stops before assigning time slots.
ROUTE-03 Medium	`itinerary-detail.page.ts` L806–842; `suggested-wizard.page.ts` L422–460	**Fake Bongabon coordinates (`15.63, 121.14`) & unpersisted Nearby stops.** `NearbyPlaceItem` lacks `latitude/longitude`, so `addNearbyPlaceToActiveDay()` injects `15.63, 121.14` for every town and mutates `currentDay.items` in memory without calling `ItineraryService.addStop()`.	Add verified coordinates to `NearbyPlaceItem` (or resolve municipality center coordinates), and persist added stops via `this.itineraryService.addStop()` / `updateItinerary()` so they survive page reloads.
ROUTE-04 Medium	`app.routes.ts`; `itinerary-detail.page.ts` L944; `map.page.ts` L719	**Guard gaps, hardcoded back navigation & non-GPS recenter button.** `/itinerary/:id` lacks `canDeactivate: [pendingChangesGuard]` for its in-place editor; back buttons hardcode `/tabs/history` or `/tabs/home` instead of checking navigation history; and the `locate-outline` button only resets camera bounds instead of reading device GPS.	Attach `pendingChangesGuard` to `/itinerary/:id`, use `IonRouterOutlet.canGoBack()` / `Location.back()` with fallback, sync map filters to URL `queryParams`, and add a real GPS *Locate Me* control alongside a separate *Reset Province View* button.
ROUTE-05 Medium	`commute-guide.data.ts` L10–139	**Static Cabanatuan-only commute guide for all legs.** Only 8 of 32 municipalities have specific entries, and every stop modal advises departing from *Cabanatuan City Central Terminal*—even when travelling 300m between two stops in Bongabon.	Make commute guidance leg-aware (`getLegCommuteInfo(prevStop, currentStop)`): detect intra-municipality hops (≤ 3 km → local tricycle/walk), inter-municipality corridor transfers, and provincial hub departures across all 32 municipalities.

4. Quantitative Spatial Analysis of Current vs. Proposed Route Clustering

To quantify the impact of `ROUTE-02`, we computed the exact great-circle Haversine leg distances across the coordinates defined in `Project/src/app/core/data/offline-places.data.ts`. While Day 1 (Bongabon) is tightly clustered at `1.51 km` across four stops, Day 2 chains *Farm Villaronte* (San Jose City, `15.7850, 120.9980`) → *Casa Batik* (Cabanatuan City, `15.4850, 120.9680`, `33.51 km` south) → *D' Pavilion Farm* (Bongabon, `15.6322, 121.1447`, `25.02 km` east) for `58.54 km`. Day 3 chains *Minalungao National Park* (General Tinio, `15.2981, 121.1223`) → *Pantabangan Dam* (`15.8126, 121.1098`, `57.23 km` north) → *Tagpuan sa Lumang Gapan* (Gapan City, `15.3065, 120.9461`, `58.94 km` south) for `116.17 km` (over 160 km on provincial highways).

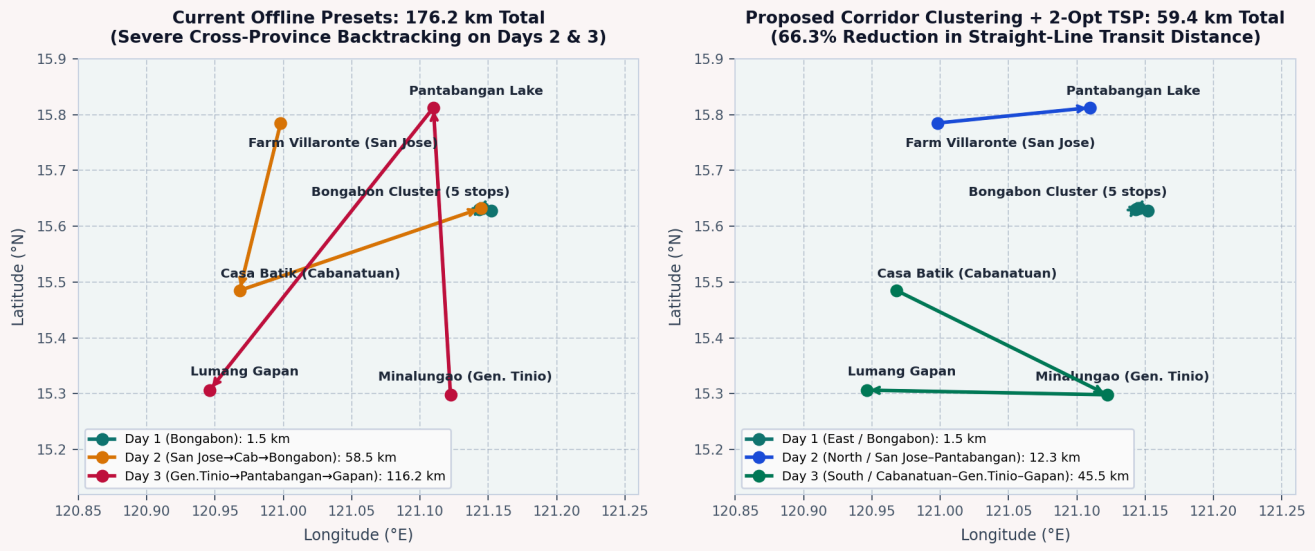


Figure 1: Coordinate plot of the 10 curated stops in `offline-places.data.ts`. Left: Current Day 1–3 presets totaling 176.2 km straight-line distance due to North-South zig-zagging. Right: Re-clustering the identical stops into East (Bongabon), North (San Jose–Pantabangan), and South (Cabanatuan–General Tinio–Gapan) corridors with 2-opt TSP ordering cuts transit distance by 66.3% to 59.4 km.

5. Proposed Target Architecture & Engineering Blueprint

In strict compliance with `ADR-001` (Zero-Paid-API & Open-Data Compliance) and W3C WCAG 2.2 accessibility standards, the proposed improvements require zero paid API keys while transforming the app from a static place list into a route-aware travel planner.

Proposal 1: Fix Leaflet MarkerCluster ESM Initialization & Resilient Layer Fallback (Fixes `MAP-01`)

In `Project/src/app/tabs/map/map.page.ts`, unwrap the Leaflet namespace object before initializing `markerClusterGroup`, attach it to `window.L` prior to plugin invocation, and fall back to a native `L.layerGroup()` so a plugin bundling mismatch can never blank out map markers:

```
// Project/src/app/tabs/map/map.page.ts
const LeafletLib: typeof L = ((L as unknown as { default?: typeof L }).default ?? L);
if (typeof window !== 'undefined') {
  (window as unknown as { L: typeof L }).L = LeafletLib;
}

// Inside initMap():
const clusterFactory = (LeafletLib as unknown as { markerClusterGroup?: (opts?: unknown) => L.MarkerClusterGroup
}).markerClusterGroup
  ?? ((window as unknown as { L?: { markerClusterGroup?: (opts?: unknown) => L.MarkerClusterGroup }
}).L?.markerClusterGroup);

this.markerCluster = typeof clusterFactory === 'function'
  ? clusterFactory({ showCoverageOnHover: false, maxClusterRadius: 40, spiderfyOnMaxZoom: true, chunkedLoading:
true })
  : (LeafletLib.layerGroup() as unknown as L.MarkerClusterGroup);
this.map.addLayer(this.markerCluster);
```

Proposal 2: Interactive Multi-Stop Route Map, Road Polylines & Multi-Waypoint Handoff (Fixes `ROUTE-01`)

- **Shared `ItineraryRouteMapComponent` & Deep-Link Mode on `/tabs/map`:** Embed a collapsible interactive Leaflet route map at the top of `ItineraryDetailPage`, `SuggestedWizardPage` (Step 2), and `CustomBuilderPage` (Timeline view), plus a *"Open Full Route on Map"* action navigating to `/tabs/map?itineraryId={id}&day={dayNumber}`.
- **Numbered Stop Pins & Road-Network Polylines:** Render numbered circular markers (`1`, `2`, `3`, with a bed icon for overnight stays) connected by directional polylines. Query the free public OSRM Demo/Community API (`https://router.project-osrm.org/route/v1/driving/{coords}?overview=full&geometries=geojson`) with a 2.5-second timeout and local IndexedDB cache; if offline or rate-limited, automatically fall back to rendering dashed geodesic Catmull-Rom spline arcs with road-winding-adjusted distance estimates (`d_road ≈ 1.28 × d_haversine` for lowland highways, `1.45 × d_haversine` for Sierra Madre mountain legs).
- **Multi-Stop Google Maps URL Builder:** Add a *"Navigate Entire Day in Google Maps"* button that constructs a complete multi-stop URL using the official Google Maps Universal Cross-Platform syntax: `https://www.google.com/maps/dir/?api=1&origin={stop1Lat},{stop1Lng}&destination={lastLat},{lastLng}&waypoints={stop2Lat},{stop2Lng}|{stop3Lat},{stop3Lng}&travelmode=driving`.

Proposal 3: Corridor-First Clustering & Intra-Day 2-Opt TSP Optimizer (Fixes `ROUTE-02`)

- **Decouple Dining vs. Activity Spatial Ordering:** In `backend/app/algorithms/itinerary_generator.py`, instead of letting the post-selection chronological sort (`DEF-03`) scramble geographic chaining, partition candidate selection into: (a) **Geographic Corridor Selection** per day (K-medoids / municipality corridor partitioning so multi-day trips progress smoothly across adjacent corridors without returning to already-visited corridors), and (b) **Intra-Day 2-Opt TSP Ordering** across non-accommodation stops, selecting the lunch dining venue nearest to the midpoint between the morning and afternoon activity stops, and pinning overnight lodging to the final evening slot (`17:30`).
- **Fix Offline Day Presets (`offline-places.data.ts`):** Update `getOfflineDayClusterPresets()` so Day 1 covers the **Bongabon Corridor** (`1.5 km`), Day 2 covers the **San Jose & Pantabangan Northern Corridor** (*Farm Villaronte* → *George Point* → *Pantabangan Dam Viewpoint* → *Highland Bali Villas*, `13.8 km`), and Day 3 covers the **Cabanatuan, General Tinio & Gapan Southern Corridor** (*Casa Batik* → *Minalungao National Park* → *Tagpuan sa Lumang Gapan*, `45.5 km`).

Proposal 4: Full-Stack Spatial & Attribute Filtering + True User Geolocation (Fixes `MAP-02`, `MAP-03`, `ROUTE-03`)

- **Backend Filter Parity (`backend/app/api/v1/places.py`):** Extend `GET /api/v1/places` to accept `municipality_name: str | None` (case-insensitive `Municipality.name.ilike`) and `price_tier: Literal['all', 'free', 'budget', 'mid'] | None` so filtering happens in SQL before `LIMIT 150` is applied. Mirror the exact same bounding-box, municipality, and price predicates inside `PlaceService.getFallbackPaginatedPlaces()`.
- **Deterministic Pin Selection on Deep-Link (`map.page.ts`):** Pass `placeId` from `HomePage.viewDestinationOnMap()` (`/tabs/map?placeId=dest-rohi-paperflower&lat=15.6315&lng=121.1440`). In `MapPage`, set a `isProgrammaticFly = true` flag during `flyTo()` so `onMapMoveEnd()` does not overwrite markers mid-flight, and automatically set `this.selectedPlace` to open the bottom sheet.
- **Real Geolocation + Separate Reset View Control:** Split the floating map control into two clear buttons: **Locate Me** (`locate-outline`, invoking `navigator.geolocation.getCurrentPosition` with a pulsing blue user dot and distance-to-POI badges if within Nueva Ecija) and **Reset Province View** (`map-outline`, flying back to `NUEVA_ECIJA_CENTER`).
- **Verified Coordinates & Persistence for Nearby Places:** Add `latitude` and `longitude` to `NearbyPlaceItem` in `offline-places.data.ts`, remove the hardcoded `15.63, 121.14` fallback in `HomePage`, `ItineraryDetailPage`, and `SuggestedWizardPage`, and persist `addNearbyPlaceToActiveDay()` via `this.itineraryService.updateItinerary()`.

Proposal 5: Angular Router Hardening & Origin-Aware Commute Guide (Fixes `ROUTE-04`, `ROUTE-05`)

- **Navigation History & Guard Parity:** Attach `canDeactivate: [pendingChangesGuard]` to `/itinerary/:id` in `app.routes.ts` so unsaved in-place edits trigger a confirmation alert. Replace hardcoded `this.router.navigate(['/tabs/history'])` in `onBack()` with an injected `IonRouterOutlet` / `Location` check (`if (this.routerOutlet?.canGoBack()) { this.navCtrl.back(); } else { this.router.navigate(['/tabs/history']); }`). Add a root session guard redirecting returning authenticated/guest users from `/` directly to `/tabs/home`, and add `/tabs/saved` as a route alias for `/tabs/history`.
- **URL Query-Parameter Filter Synchronization:** Sync active map search, category, municipality, and price filters to URL query parameters via `router.navigate([], { relativeTo: this.route, queryParams, queryParamsHandling: 'merge', replaceUrl: true })` so filtered map states survive browser refreshes and link sharing.

- **Leg-Aware Provincial Commute Engine (`commute-guide.data.ts`):** Upgrade `getCommuteInfoForMunicipality(municipalityName, placeName, previousStop?)` to compare the previous stop's coordinates and municipality against the current stop: if both stops are in the same municipality and ≤ 1.2 km apart, recommend a *5–10 min walk or local tricycle hop (₱15–₱25)*; if in the same municipality (> 1.2 km), recommend a *local tricycle transfer*; if crossing between adjacent towns on the same highway corridor (e.g., Gapan → General Tinio or San Jose → Pantabangan), show the direct corridor jeepney/UV Express transfer rather than routing back through Cabanatuan City.

6. Phased Implementation Roadmap & Verification Plan

The proposed improvements are structured into three low-risk, incrementally verifiable engineering phases that preserve backwards compatibility across existing REST contracts and unit tests.

Phase	Scope & Target Files	Key Deliverables	Verification Gate
Phase 1 Immediate Bug Fixes (Day 1)	`map.page.ts`, `place.service.ts`, `places.py`, `home.page.ts`, `offline-places.data.ts`	Fix `MAP-01` ESM `markerClusterGroup` initialization; fix `MAP-02` server/offline bounds, municipality, and price filtering; fix `MAP-03` deep-link bottom-sheet selection; fix `ROUTE-02` offline preset zig-zag and `ROUTE-03` nearby coordinates.	`npm run test:unit` (add unmocked `initMap()` pin DOM test); Playwright E2E assertion that `.custom-map-pin` count > 0 on `/tabs/map`.
Phase 2 Route Map & TSP Engine (Days 2–3)	`routing.service.ts` (new), `itinerary-route-map.component.ts` (new), `itinerary_generator.py`	Build reusable multi-stop route map with numbered pins, OSRM road polylines + offline geodesic spline fallback, leg distance/time pills, multi-waypoint Google Maps URLs, and 2-opt TSP stop ordering.	`pytest backend/tests/test_itineraries.py` asserting intra-day distance monotonicity and 0 cross-corridor backtracking.
Phase 3 Router Polish & Leg Commute (Day 4)	`app.routes.ts`, `tabs.routes.ts`, `itinerary-detail.page.ts`, `commute-guide.data.ts`	Add `pendingChangesGuard` to `/itinerary/:id`, history-aware back navigation, URL query-param sync on `/tabs/map`, GPS *Locate Me* control, and leg-aware 32-municipality commute guide.	Full CI suite: `npm run lint`, `npm run build` (0 budget warnings), `npm run test:e2e`.